32)Bit stuffing Mechanism using C

Program:

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#include <stdio.h>

void bitStuffing(int data[], int n) {
    int stuffed_data[100];
    int i, j = 0, count = 0;

    for (i = 0; i < n; i++) {
        stuffed_data[j] = data[i];
        j++;

        if (data[i] == 1) {
            count++;
        } else {
            count = 0;
        }

        // If five consecutive 1s are encountered, stuff a 0
        if (count == 5) {
            stuffed_data[j] = 0;
            j++;
            count = 0; // Reset counter after stuffing
        }
    }

    printf("Data after Bit Stuffing: ");
    for (i = 0; i < j; i++) {
        printf("%d", stuffed_data[i]);
```

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    }
    printf("\n");
}

void bitDestuffing(int stuffed_data[], int stuffed_len) {
    int destuffed_data[100];
    int i, j = 0, count = 0;

    for (i = 0; i < stuffed_len; i++) {
        destuffed_data[j] = stuffed_data[i];
        j++;

        if (stuffed_data[i] == 1) {
            count++;
        } else {
            count = 0;
        }

        // If five consecutive 1s are encountered, skip the next stuffed 0
        if (count == 5) {
            i++; // Skip the next bit (which is the stuffed 0)
            count = 0; // Reset counter
        }
    }

    printf("Data after Bit De-stuffing: ");
    for (i = 0; i < j; i++) {
        printf("%d", destuffed_data[i]);
    }
```

```
    printf("\n");
}

int main() {
    int n, stuffed_len;
    int data[100], stuffed_data[100];
    int i;

    // --- Bit Stuffing Section ---
    printf("Enter number of bits: ");
    scanf("%d", &n);

    printf("Enter the bits (0/1):\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &data[i]);
    }

    bitStuffing(data, n);

    // --- Bit De-stuffing Section ---
    printf("\nEnter stuffed data length: ");
    scanf("%d", &stuffed_len);

    printf("Enter the stuffed bits (0/1):\n");
    for (i = 0; i < stuffed_len; i++) {
        scanf("%d", &stuffed_data[i]);
    }

    bitDestuffing(stuffed_data, stuffed_len);
```

```
    return 0;
}
```

Output: